

Project – Credit scorecard

Data: `scorecard_dataset.xlsx`

It consists of 10 sheets.

Name	Description
IntRate	Interest rate assigned to the loan (linear model y). Companion binary y for the logistic model: default indicator (1 = defaulted).
Income	Borrower gross annual income
DTI	Debt-to-income ratio (monthly debt payments / monthly income)
LoanAmt	Principal amount of the loan
HistLen	Length of borrower's credit history
Utilisation	Revolving credit utilisation (balance / credit limit)
Age	Borrower age
Offences	Number of driving or other offences
EmpStatus	Employment status of borrower
Purpose	Purpose of the loan
HomeOwn	Housing status of borrower
Gender	Male/Female
Education	Level of education of applicant/borrower

Lenders must decide which loan applicants to approve and what interest rate to charge, but default risk varies widely across borrowers. Mispricing this risk leads either to losses (rates too low for risky borrowers) or lost business (rates too high for safe ones). Using cross-sectional loan-level data, you are required to build a two-part credit model:

- A model explaining the interest rate assigned (to test whether observed pricing is consistent with modelled risk).
- A model estimating each borrower's probability of default (PD). (Here, the Interest Rate may be used as an independent variable.)